THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · NCERT-TYPE UNIT TEST SET 2

Chapters 1–4: **Coordinates, Linear Polynomials, World of Numbers, Algebraic Identities** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. By identity, $103^2 =$ 1

(a) 10611 **(b)** 10609 **(c)** 10608 **(d)** 10610

2. If $x < 0$ and $y > 0$, the point (x, y) lies in: 1

(a) Quadrant IV **(b)** Quadrant III **(c)** Quadrant I **(d)** Quadrant II

3. The graph of $y = 2x - 8$ cuts the x-axis at: 1

(a) $(5, 0)$ **(b)** $(4, 0)$ **(c)** $(0, 4)$ **(d)** $(-4, 0)$

4. A rational number $\frac{p}{q}$ equal to $0.1\overline{6}$ is: 1

(a) $\frac{1}{3}$ **(b)** $\frac{1}{7}$ **(c)** 0 **(d)** $\frac{1}{6}$

5. $x^2 + 8x + 16$ is the square of: 1

(a) $(x - 4)$ **(b)** $(x + 4)$ **(c)** $(x + 16)$ **(d)** $(x + 8)$

6. The distance of $(5, -7)$ from the x-axis is: 1

(a) 9 **(b)** 7 **(c)** 8 **(d)** 6

7. The sum of the zeros of $2x - 6$ and $5x - 20$ is: 1

(a) 8 **(b)** 6 **(c)** 7 **(d)** 9

8. $(-15) - (-23) =$ 1

(a) 7 **(b)** 9 **(c)** 10 **(d)** 8

9. Factorise $x^2 + 7x + 10$: 1

(a) $(x + 4)(x + 5)$ **(b)** $(x - 2)(x - 5)$ **(c)** $(x + 3)(x + 4)$ **(d)** $(x + 2)(x + 5)$

10. If the midpoint of AB is $(5, 7)$ and $A = (2, 3)$, then B is: 1

(a) $(8, 11)$ **(b)** $(-8, 11)$ **(c)** $(8, -11)$ **(d)** $(11, 8)$

11. The nth term of the pattern 1, 7, 13, 19, ... is: 1

(a) $6n - 6$ **(b)** $7n - 5$ **(c)** $6n - 5$ **(d)** $6n - 4$

12. Which of the following has a terminating decimal expansion? 1

(a) $\frac{2}{7}$ **(b)** $\frac{5}{6}$ **(c)** $\frac{1}{3}$ **(d)** $\frac{7}{16}$

13. If $x + y = 9$ and $xy = 14$, then $x^2 + y^2 =$ 1

(a) 52 **(b)** 55 **(c)** 53 **(d)** 54

14. The point $(-3, 8)$ is reflected in the y-axis. The image is: 1

(a) $(3, -8)$ **(b)** $(8, 3)$ **(c)** $(-3, 8)$ **(d)** $(3, 8)$

15. A plumber charges ₹100 plus ₹50 per hour. For 3 hours he charges ₹: 1

(a) 251 **(b)** 250 **(c)** 249 **(d)** 252

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. Find $p(2)$ and $p(-2)$ for $p(x) = 4x + 3$. 2

17. Write $\frac{1}{6}$ as a decimal. 2

18. Find 104^2 using an identity. 2

19. The midpoint of AB is $M(4, 6)$ and $A = (1, 2)$. Find B. 2

20. Write the slope and intercept of $y = 8x + 3$, and the coordinates of the point where it cuts the y-axis. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Factorise $x^2 + 8x + 12$. 3

22. A rectangle has vertices $A(1, 1)$, $B(22, 1)$, $C(22, 29)$ and D. Find D and the length of diagonal AC. 3

23. If $p(x) = 2x + 3$ and $q(x) = 3x + 7$, find $p(x) + q(x)$ and its zero. 3

24. Find three rational numbers between $\frac{1}{3}$ and $\frac{2}{3}$. 3

OR

Evaluate 103×107 using $(x + a)(x + b)$. 3

25. Show that the points $A(1, 1)$, $B(5, 4)$ and $C(9, 1)$ are the vertices of an isosceles triangle. 3

OR

Draw up a table for $y = 3x - 1$ for $x = -1, 0, 1, 2$ and state the slope and the y-intercept. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. The vertices of a triangle are $A(0, 0)$, $B(-9, 12)$, $C(-9, 0)$. Find the lengths of its three sides and its perimeter. 5

OR

For $y = 4x - 12$: 5

(i) find y at $x = 0, 1, 2$;

(ii) find the point where the graph crosses the x-axis;

(iii) state the slope and say whether the graph rises or falls from left to right.

27. Simplify $(5 + \sqrt{3})(5 - \sqrt{3})$ and $(5 + \sqrt{3})^2$ and say which result is rational. 5

OR

Expand $(x + 2)^3$. 5

END OF QUESTION PAPER — **Total: 50 Marks**